

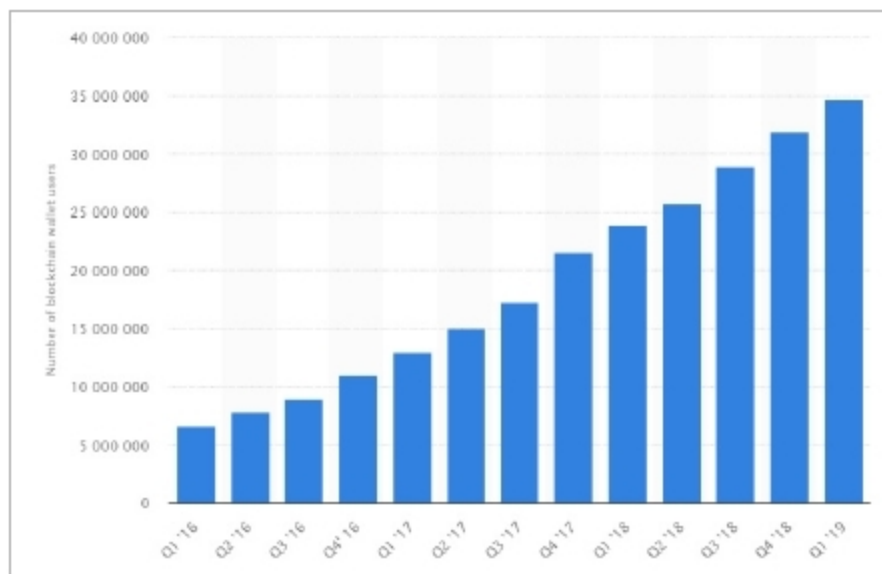


Figure 1: Number of blockchain wallet users worldwide (2016-2019)
[Source: Statista, the statistics portal]

data elements and records. Each block participates in the blockchain network and is immutable. The term ‘immutable’ here means that it is secured and non-breakable. Hence, it forms the blockchain of a secured chain of blocks without any probabilities of intentional or accidental tampering and leakage of data.

The first block in the chain of networks is known as the Genesis Block from where the blockchain initiates the transactions. The chain grows by inserting different blocks. Every block is encrypted with the previous block, and hence becomes secure. It is difficult to crack the previous states because of so many encryptions.

Tools and frameworks for blockchain development

Framework/Tool	URL
Remix IDE	http://remix.ethereum.org/
EthFiddle	https://ethfiddle.com/
Truffle	https://truffleframework.com/
Embark	https://embark.status.im/
Populus	http://populus.readthedocs.io/en/latest/
Go Ethereum/Geth	https://geth.ethereum.org/ https://github.com/ethereum/go-ethereum/wiki/geth

For information on frameworks like Etherlime, Dot-Abi-cli, Web3JS, PyEthereum, Nethereum, Cava, Solidity, CPP Ethereum, Liquidity, Infura, Lamden, MyThrill, Coq, etc, do go online.

Python tools for blockchain programming

Python is a key programming language that is used in almost every area of high performance computing. It provides the tools and libraries that can be used for blockchain development including decentralised applications. As in blockchain technology, there are secured protocols and algorithms, Python has numerous toolkits available on its official repository <https://pypi.org/>. The specific libraries for decentralised applications and blockchain implementations

are available at the following URLs:

- <https://pypi.org/search/?q=blockchain>
- <https://pypi.org/search/?q=dapp>

Installation of Python tools and libraries for blockchain programming

The additional libraries and toolkits can be installed with existing Python using the Pip installer, as follows:

```
$ pip install <packagename>
```

In the Windows operating system, the following instruction is used:

```
WindowsDrive:\Python Installation Directory>python -m pip  
install <packagename>
```

The blockchain technology is highly dependent on and makes use of integration with dynamic cryptography and encryption. For this, the Hashlib library can be installed using the above instruction.

Building blockchain applications

The following scenario is of a secured blockchain that is generating hash values so that overall transactions and records are highly secured. In the following code *blockchainhash.py*, the dynamic hash value is generated. This is the base of any blockchain with different transactions in a chain, which makes the overall blockchain.

```
blockchainhash.py
import datetime as date
import hashlib as hasher

class Block:
    def __init__(self, idx, ts, mydata, backhash):
        self.idx = idx
        self.ts = ts
        self.mydata = mydata
        self.backhash = backhash
        self.hash = self.hashop()

    def hashop(self):
        shahash = hasher.sha256()
        shahash.update(str(self.idx) + str(self.ts) + str(self.
mydata) + str(self.backhash))
        return shahash.hexdigest()

def genesis():
    return Block(0, date.datetime.now(), "Genesis Block", "0")

def next_block(last_block):
    this_idx = last_block.idx + 1
    this_ts = date.datetime.now()
    this_mydata = "Block" + str(this_idx)
    this_hash = last_block.hash
    return Block(this_idx, this_ts, this_mydata, this_hash)

blockchain = [genesis()]
```